

Networking & Troubleshooting:

Basics of Computer Networks

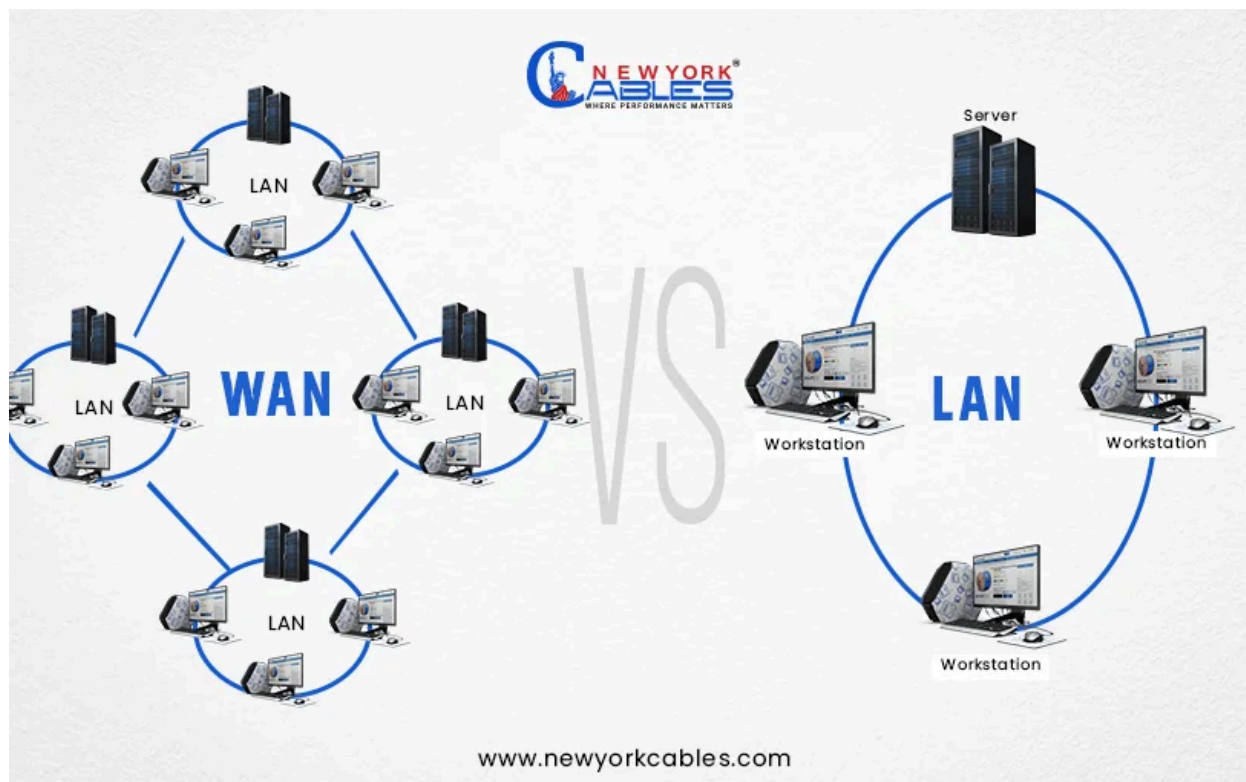
LAN (Local Area Network)

- Small area network (e.g., home, office, school).
- Connects devices within a limited space (building or campus).
- Fast speed, low cost, usually wired (Ethernet) or wireless (Wi-Fi).
- Example: Computers + printer in one room connected via a router.

WAN (Wide Area Network)

- Large area network connecting multiple LANs over long distances (cities, countries).
- Slower than LAN, higher cost.
- Internet is the largest WAN.
- Example: Branch offices connected across India.

Here are clear diagrams comparing **LAN vs WAN** (showing coverage area, speed, and examples):



newyorkcables.com

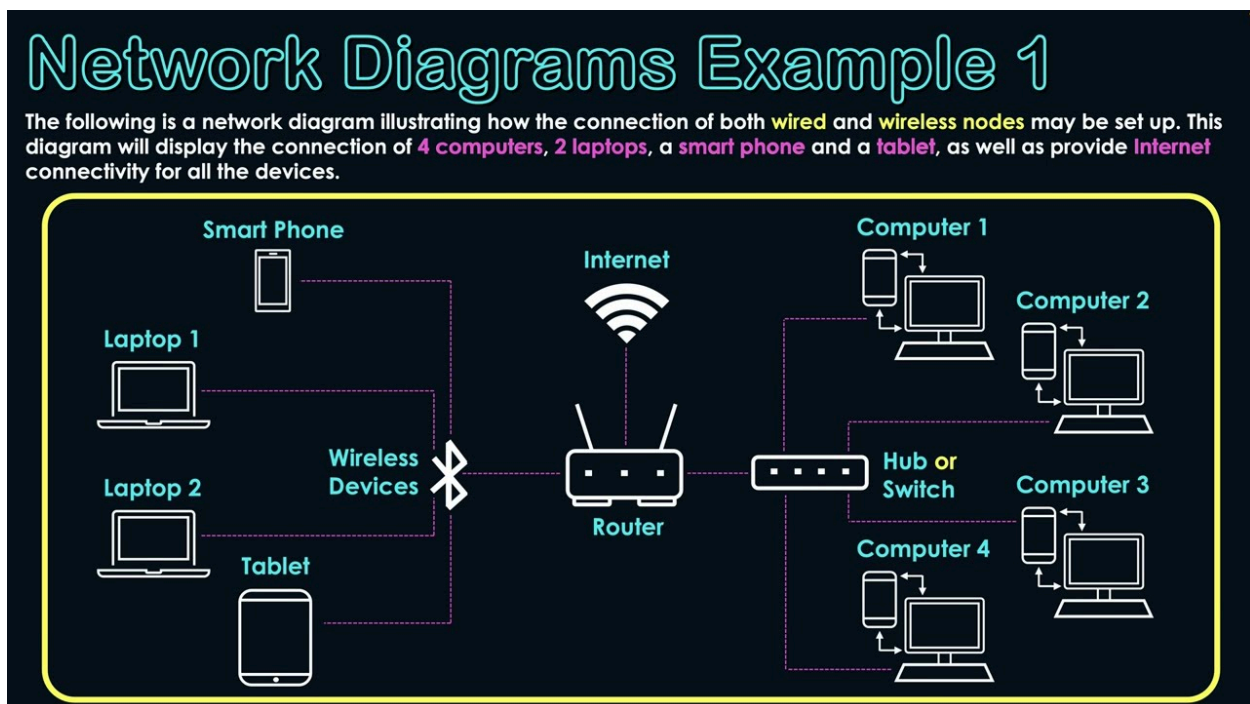
LAN vs WAN vs VLAN			
	LAN	WAN	VLAN
Geographical area	Small	Global	Medium
Good for	Schools, colleges, offices, small businesses	Large or global enterprises	Medium-to-large businesses
Technical requirements	Ethernet and switches; L3 switching and routing	MPLS, VPLS, satellite technologies	Ethernet and switches

unitedworldtelecom.com

Wi-Fi

- Wireless networking technology (using radio waves).
- Allows devices to connect without cables (laptops, phones, smart TVs).
- Range: 20-100 meters indoors (depends on router).
- Uses a **Wi-Fi router** to create a hotspot.

Here are simple diagrams showing **how Wi-Fi works** (router broadcasting to devices):



[youtube.com](https://www.youtube.com/watch?v=7H8v8v8v8v8)

Network Diagram Example 1: Wired & Wireless Connections

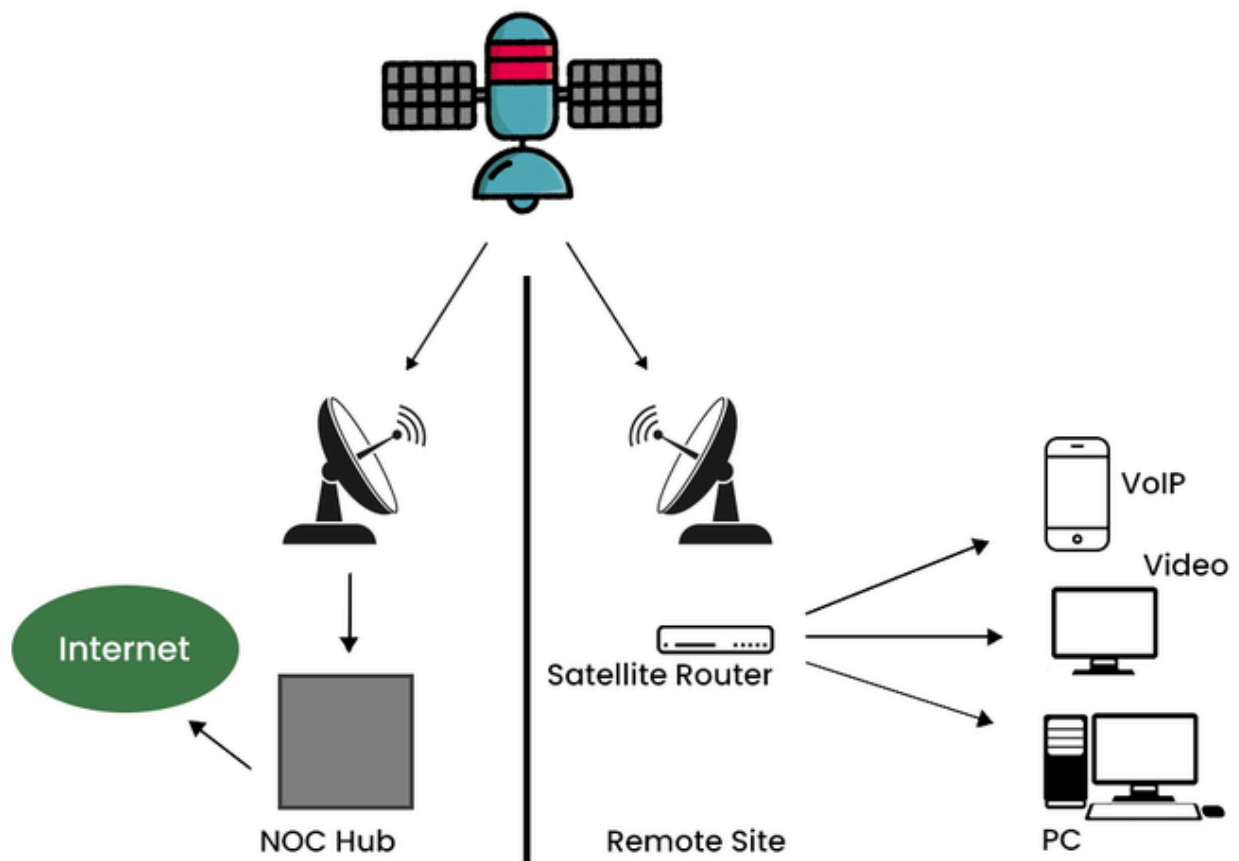
Broadband

- High-speed internet connection (faster than dial-up).
- Types: Fiber optic, DSL, Cable, 4G/5G mobile.
- Always-on, no need to dial every time.

Bluetooth

- Short-range wireless technology (up to 10 meters).
- Used for connecting headphones, mouse, keyboard, file transfer between phones.
- Low power, low speed compared to Wi-Fi.

Here are examples of common **internet connection types** (icons for dial-up, broadband, Wi-Fi):



geeksforgeeks.org

Types of Internet Connection - GeeksforGeeks

Internet: Concept, Applications, Connecting, Troubleshooting

Internet is a global network of networks connecting billions of devices worldwide.

Applications of Internet:

- Browsing websites
- Email communication
- Online shopping/banking
- Video streaming (YouTube, Netflix)
- Social media
- Online education
- Gaming
- Cloud storage

Connecting to the Internet:

1. Via ISP (Internet Service Provider) like Airtel, Jio, BSNL.
2. Methods: Broadband (most common), Mobile data (4G/5G), Wi-Fi hotspot.

Troubleshooting common issues:

- Restart modem/router/computer.
- Check cables/Wi-Fi connection.
- Forget and reconnect Wi-Fi.
- Run network troubleshooter (Settings → Network & Internet → Status → Network troubleshooter).

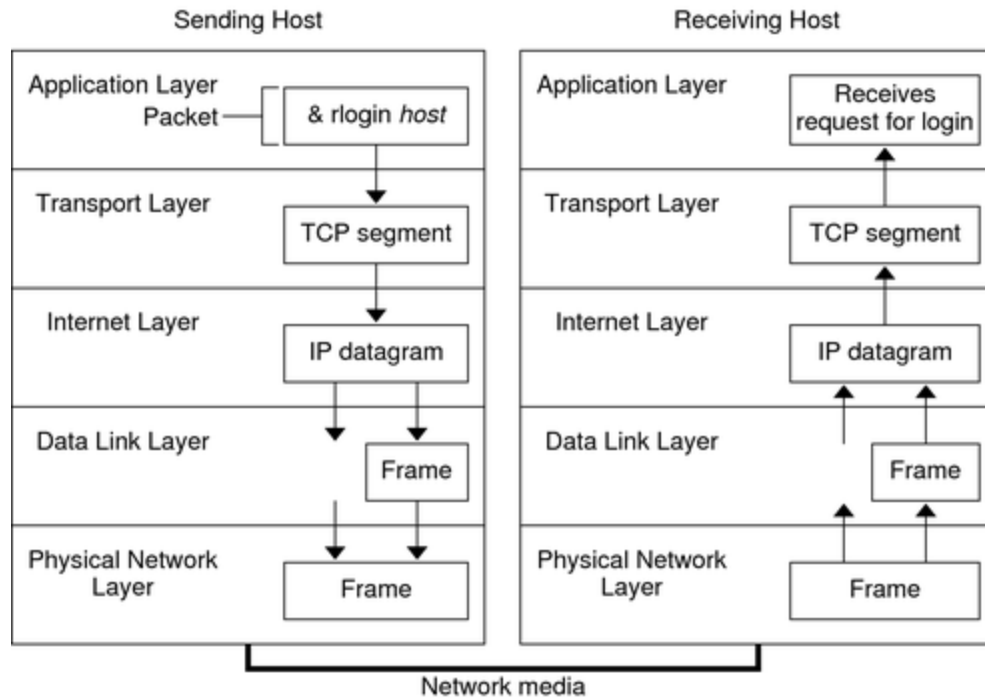
WWW, TCP/IP, DNS, Search Engine

WWW (World Wide Web) — System of interlinked web pages accessed via internet (not the same as internet).

TCP/IP — Suite of protocols (rules) for communication on the internet.

Main layers: Application, Transport, Internet, Link.

Here are standard **TCP/IP model layers** diagrams:

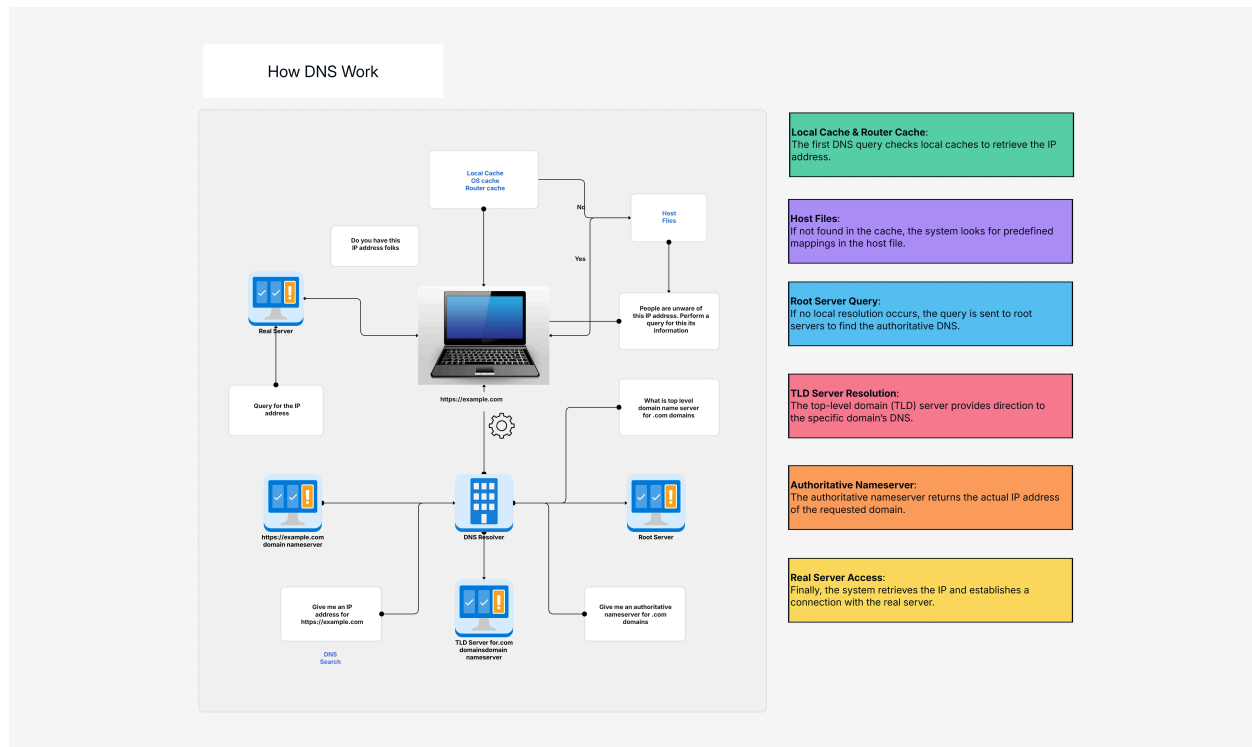


docs.oracle.com

How the TCP/IP Protocols Handle Data Communications - Oracle ...

DNS (Domain Name System) — Translates human-friendly names (google.com) to IP addresses (142.250.190.174).

Here are flowcharts showing **how DNS works** (query from browser to DNS server):



clouddairy.com

How DNS Works: A Simple Guide To Internet Name Resolution

Search Engine — Tool to find information (Google, Bing, DuckDuckGo). Type keywords → get results.

Key Web Browser Features

Modern browsers (Chrome, Edge, Firefox) include:

- Address bar (URL entry)
- Tabs (multiple pages)
- Bookmarks/Favorites
- History
- Incognito/Private mode
- Extensions/Add-ons
- Password manager
- Zoom, dark mode

Here are labeled screenshots of a **modern web browser** interface:



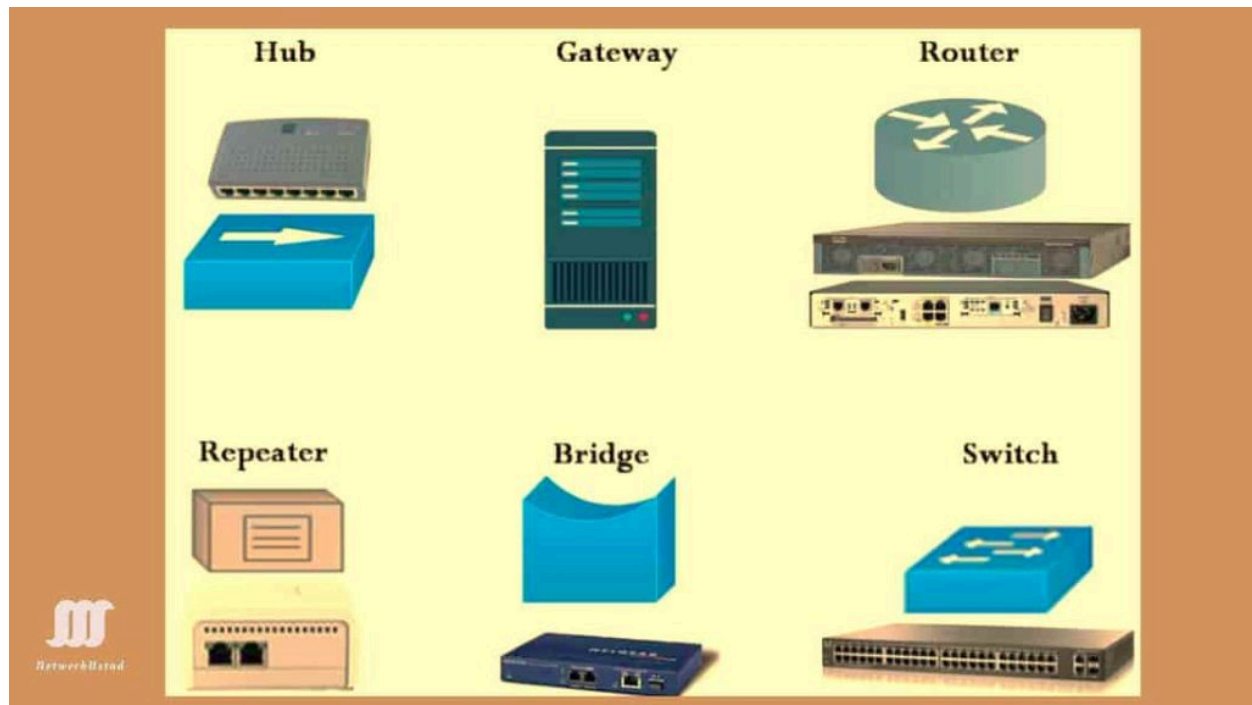
[techsnail.com](https://www.techsnail.com)

What is a web browser?

Brief about Switch, Router, Gateway

- **Switch** → Connects devices in a LAN (fast, local traffic).
- **Router** → Connects LAN to internet/WAN, routes traffic, provides Wi-Fi.
- **Gateway** → Entry/exit point between networks (often router acts as gateway).

Here are comparison diagrams of **network devices** (switch, router, gateway):



networkkustad.com

Network Devices: A Complete Guide to Understanding and Choosing ...

Various Applications of Internet

- **E-mail** — Send/receive messages (Gmail, Outlook).
- **Information gathering** — Research via search engines.
- **Retailing** — E-commerce (Amazon, Flipkart).
- Others: Online banking, government services (Aadhaar, income tax), telemedicine.

Methods of Connecting to the Internet

- **Dial-up** — Old, slow (phone line + modem).
- **ISDN** — Faster digital version of dial-up (rare now).
- **Broadband** — High-speed, always-on (fiber, cable, DSL).

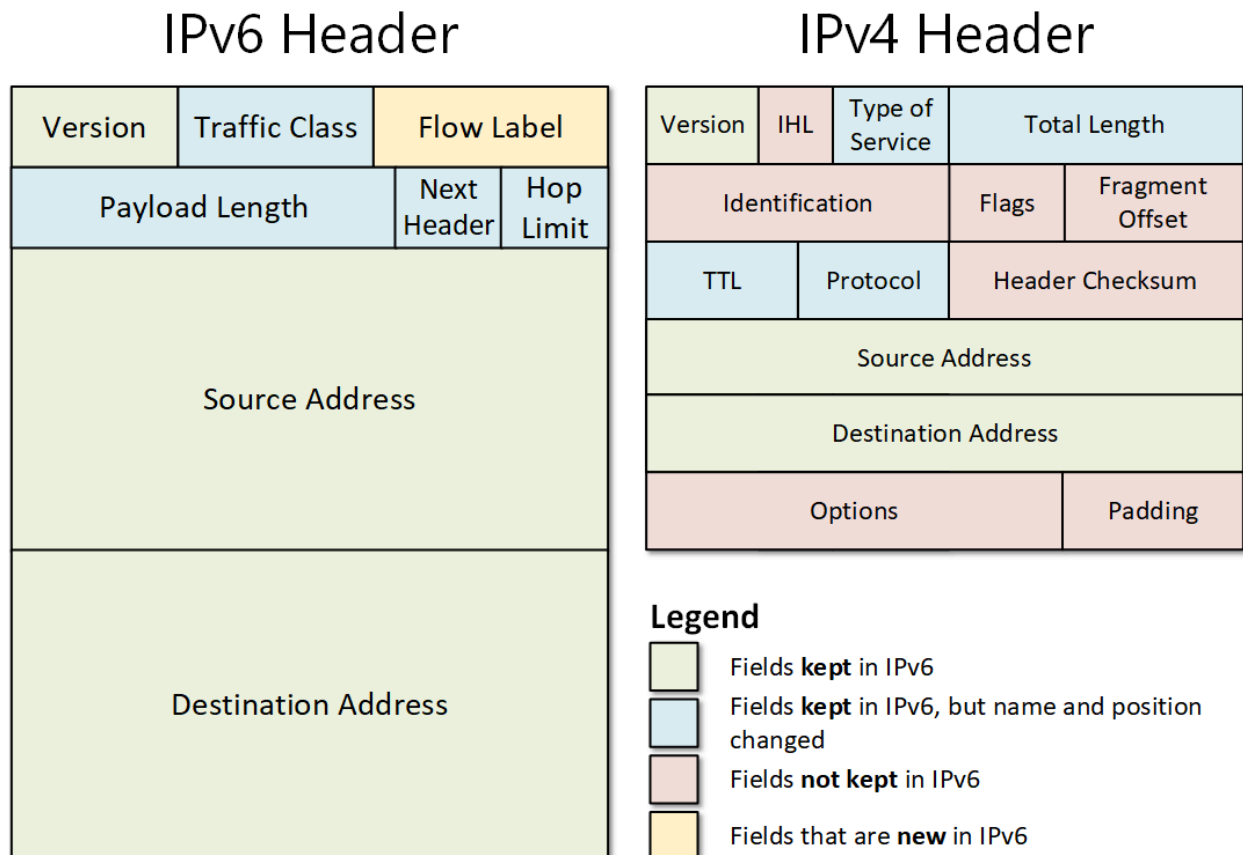
Brief Introduction to Internet Addressing, Protocols

- **Internet Addressing** — Every device has an IP address (IPv4: 192.168.1.1 or IPv6: longer hexadecimal).
- **Protocols:**
 - **TCP/IP** → Core internet protocol.
 - **FTP** → File Transfer Protocol (uploading/downloading files).
 - **HTTP/HTTPS** → HyperText Transfer Protocol (secure for web pages).

IPv4 vs IPv6:

- IPv4: 32-bit, ~4 billion addresses (running out).
- IPv6: 128-bit, huge number of addresses (future standard).

Here are comparison visuals of **IPv4 vs IPv6** addresses:



networkacademy.io

Define and Understand the Terms

- **Internet Service Provider (ISP)** — Company providing internet access (e.g., Jio, Airtel).
- **Uniform Resource Locator (URL)** — Web address (e.g., <https://www.google.com>).
- **Hyperlink** — Clickable text/link that takes you to another page/resource.

Overview of Use of Search Engines and E-mail Messages

- **Search Engines** — Enter keywords → results with titles, snippets, links.
- **E-mail** — Compose → To: address → Subject → Body → Send. Attachments possible.

Instant Messaging and Collaboration

- **Instant Messaging** — Real-time text chat (WhatsApp, Telegram, Signal).
- **Providers** — WhatsApp (most popular in India), Telegram, Microsoft Teams (for work).

Use of Social Networking Sites

- **Facebook** — Connect with friends, share posts/photos, groups.
- **Twitter (now X)** — Short messages (posts), trends, news.
- Others: Instagram (photos/videos), LinkedIn (professional).

Network Troubleshooting

Basic steps:

1. Restart devices (PC, router, modem).
2. Check physical connections (cables/Wi-Fi).
3. Verify IP address (Command Prompt → ipconfig).
4. Flush DNS (ipconfig /flushdns).

5. Use built-in troubleshooter.
6. Contact ISP if no internet.